

Translated

From: e-maxx.ru

Submask Enumeration

Enumerating all submasks of a given mask

Given a bitmask m , you want to efficiently iterate through all of its submasks, that is, masks s in which only bits that were included in mask m are set.

Consider the implementation of this algorithm, based on tricks with bit operations:

```
int s = m;
while (s > 0) {
    ... you can use s ...
    s = (s-1) & m;
}
```

or, using a more compact `for` statement:

```
for (int s=m; s; s=(s-1)&m)
    ... you can use s ...
```

In both variants of the code, the submask equal to zero will not be processed. We can either process it outside the loop, or use a less elegant design, for example:

```
for (int s=m; ; s=(s-1)&m) {
    ... you can use s ...
    if (s==0) break;
}
```

Let us examine why the above code visits all submasks of m , without repetition, and in descending order.

Suppose we have a current bitmask s , and we want to move on to the next bitmask. By subtracting from the mask s one unit, we will remove the rightmost set bit and all bits to the right of it will become 1. Then we remove all the "extra" one bits that are not included in the mask m and therefore can't be a part of a submask. We do this removal by using the bitwise operation `(s-1) & m`. As a result, we "cut" mask $s-1$ to determine the highest value that it can take, that is, the next submask after s in descending order.